

Slovak Technical University in Bratislava
Faculty of Informatics and Information Technologies

FIIT-127326

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**Unified Interoperability: Bridging
Homogeneous and Heterogeneous
Blockchains**

Bachelor Thesis

Supervisor: Ing. Ing. Dušan Morháč

Máj 2025

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Study Program: Software Engineering

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Annotation

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Author: Adam Svitek

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As the blockchain landscape continues to expand, the need for effective communication between networks becomes more critical. While some blockchains share similar architectures (homogeneous), others differ significantly in design, consensus mechanisms, and data structures (heterogeneous). Achieving interoperability across both types introduces a range of technical challenges, especially when aiming to create seamless user experiences. Layer 2 (L2) solutions and interoperability protocols attempt to address these issues by abstracting underlying complexities.

Analyze the interoperability challenges and opportunities between both homogeneous and heterogeneous blockchain networks. Design a solution or framework that enables secure and efficient interaction between diverse blockchain systems, and implement a working prototype or conceptual model. Evaluate the effectiveness and scalability of their proposed approach, discuss the implications and possible extensions of their work, and conclude with practical insights and recommendations for future research and development.

Potom, vložit finálny návrh zadania diplomovej práce

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The acknowledgements and the people to thank go here, don't forget to include your project advisor(Ing. Dušan Morháč)...

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Chapter 1

Technical abstract

Modern blockchain ecosystems are divided across multiple networks, execution layers, and interoperability mechanisms. This helps with scalability, but it also makes asset transfers more difficult for users. A transfer that looks simple from the user's point of view may require several technical steps, especially when funds need to move between Ethereum Layer 1, Ethereum Layer 2 networks, and external ecosystems such as Polkadot.

Existing interoperability protocols already solve important parts of this problem. Some protocols focus on transfers within the Ethereum ecosystem, while others enable communication between Ethereum and other blockchain networks. However, these solutions are usually provided as separate tools. In more complex cases, the user still has to choose the correct protocol, switch between different applications, and execute the required steps in the correct order. This increases the complexity of the process and creates space for mistakes.

The goal of this thesis was to design and implement a web-based application that reduces this fragmentation at the application layer. The proposed solution does

not create a new bridge or interoperability protocol. Instead, it combines existing protocols into one guided transfer workflow. One protocol is used for transfers within the Ethereum ecosystem, while other is used for transfers between Ethereum and Polkadot.

The application was implemented as a client-side solution. It communicates with protocol APIs, blockchain RPC endpoints, and the user's wallet. Based on the selected source network, destination network, asset, amount, and recipient address, the application prepares a transfer plan and executes the required steps in sequence.

The implemented solution was evaluated in testnet environments using several transfer scenarios. The results show that the application mainly improves workflow coordination rather than protocol-level transfer speed. Its main contribution is the simplification of multi-step blockchain asset transfers through a single user-facing interface.

Chapter 2

Lay summary

Moving digital assets between different blockchain networks can be difficult for ordinary users. When a user wants to send tokens or cryptocurrency from one network to another, the process is often not completed in a single place. The user may need to open several tools, choose the correct transfer path, understand which connection_bridge should be used, and approve more than one transaction in the correct order. This can be confusing, especially when the transfer goes across different ecosystems such as Ethereum and Polkadot.

This thesis focuses on simplifying this process. The result of the work is a web application that guides the user through the transfer step by step. Instead of requiring the user to manually search for the right tools and decide how the transfer should be executed, the application collects only the necessary information: the source network, the destination network, the asset, the amount, and the recipient address. Based on this input, the application prepares the required transfer steps and presents the process through one unified interface. The final transactions are still approved by the user in their own wallet, so the application does not take control of the user's funds.

The proposed application does not make the blockchain transfer itself faster, because it still depends on the existing networks and protocols. However, it can make the whole process faster for the user. The user does not have to search for the right tools, decide which bridge to use, or manually plan the next step. By combining the process into one guided workflow, the application saves time, reduces confusion, and lowers the chance of mistakes.

Chapter 3

Introduction

Blockchains over time developed from isolated decentralized networks to larger ecosystems composed of multiple layers , protocols and specialized environments. One example of this development is Ethereum, since users can not only interact with the main Layer 1 network, but also with several Layer 2 networks which were created to improve scalability and reduce higher transaction costs. At the same time, other ecosystems such as Polkadot use different architectures and provide their own way of organizing assets, applications, and communication between chains.

This ongoing development also creates a practical problem for users who need to move digital assets between these multiple environments. Although many interoperability protocols and bridges already exist, they often focus only on specific part of the transfer path. A user may be able to move assets between Ethereum Layer 2 networks and also Layer 1 network back and forth but transferring the same assets further to another ecosystem may require a different tool and another transaction process. As a result, more complex transfers are still fragmented across many different interfaces and require manual decisions from the user what can be

a setback for some.

The problem therefore is the way how the interoperability is presented and coordinated for the user. Users to make such transactions must understand which protocol should be used, which network should be selected and in what order the transactions should be executed. This increases the difficulty of the whole process and creates a higher risk of mistakes.

This thesis focuses on reducing this fragmentation at the application layer. Instead of designing a new bridge or protocol, the work proposes a web application that combines selected existing interoperability mechanisms into one guided transfer workflow. The main goal is to make existing transfer routes easier to access, simplify route selection, support transfer planning, and provide step-by-step execution for selected transfer scenarios between Ethereum-based networks and the Polkadot ecosystem.

Chapter 4

Problem statement and solution

4.1 Problem statement

The problem addressed in this thesis is the fragmentation of asset transfers across different blockchain networks, layers, and ecosystems. Some blockchain networks may share similar technical properties, such as wallet interaction, address format, smart contract execution, or token standards, while others are based on different architectures. This thesis focuses mainly on Ethereum-based networks and the Polkadot ecosystem. Ethereum consists of the main Layer 1 network and several Layer 2 networks, which are closely related to Ethereum and often share similar user interaction patterns. Polkadot, on the other hand, is a different ecosystem with a different architecture, addressing model, and communication mechanisms between chains.

Users may need to move assets across these environments, for example from an Ethereum Layer 2 network to Polkadot or in the opposite direction. Although existing interoperability solutions support many individual transfer paths, they often do not provide one complete workflow for more complex routes. Some protocols

are suitable for transfers inside the Ethereum ecosystem, while others are used for transfers between Ethereum and Polkadot. As a result, more complex transfers may require several dependent steps, correct execution order, and the use of more than one web application.

This creates several difficulties for the user. First, the user must identify the correct route, choose the correct bridge or protocol, switch between different interfaces, enter the correct recipient address format, and approve the required transactions. Each of these manual decisions increases the overall complexity of the process and creates a higher risk of mistakes.

The main problem is therefore not only technical interoperability between blockchains, but also the lack of a unified application-level workflow that coordinates selected transfer steps for the user.

4.2 Technical literature review

4.2.1 Approaches to blockchain interoperability

Blockchain interoperability has been studied and implemented through several approaches. Its main goal is to allow assets and messages to move between blockchain environments that were originally designed as independent systems. Existing solutions differentiate in the problem they solve. Some protocols focus on fast asset transfers inside one ecosystem and others focus on secure communication between heterogeneous blockchains.

Protocols optimized for speed often rely on relayers or liquidity providers, which improves user experience but introduces additional dependencies. More security-focused solutions may use light clients or cryptographic proofs, which can reduce trust assumptions but also increase complexity and execution time. Therefore, no

single protocol solves all interoperability problems in the same way.

4.2.2 Interoperability inside the Ethereum ecosystem

Inside the Ethereum ecosystem, interoperability is closely connected to the connection between Layer 1 and Layer 2 networks. Layer 2 networks improve scalability and reduce transaction costs, but they also create the need for reliable asset transfers between layers. Users may need to transfer assets from one Layer 2 network to another, or from a Layer 2 network back to Ethereum Layer 1.

Protocols such as **Across Protocol** and **Hop Protocol** address this part of the problem. Their purpose is to make transfers inside the Ethereum ecosystem faster. Across Protocol uses a relayers, where the user inputs transfer parameters and relayers provide liquidity on the destination chain. Hop Protocol' is similar to Across as they are improving the same transfer part.

Since they operate solely over the Ethereum ecosystem they are unable to transfer tokens across multiple heterogeneous blockchains.

4.2.3 Interoperability between Ethereum and Polkadot

Interoperability between Ethereum and Polkadot is more complex because both ecosystems use different architectures and execution models. Ethereum uses an account-based smart contract environment, while Polkadot is based on a relay chain and parachain architecture. Because of this, transfers between these ecosystems require different interoperability mechanisms than transfers between Ethereum Layer 2 networks.

Snowbridge is an example of a protocol focused on transfers between Ethereum and Polkadot. It uses light-client verification, which allows one ecosystem to verify

information from the other without relying only on a trusted third party. This improves the security compared to bridges based on centralized validators.

The limitation of Snowbridge is that it mainly solves the Ethereum to Polkadot part of the route. If assets start on an Ethereum Layer 2 network, they may first need to be moved to Ethereum Layer 1 before the Snowbridge transfer can be executed. Therefore, Snowbridge is suitable for an important part of the transfer path, but it does not fully coordinate the complete multi-step workflow.

4.2.4 Related interoperability solutions

Other interoperability solutions address similar problems with different trade-offs. **Celer cBridge** is a liquidity-oriented bridge focused on fast transfers across supported networks. Its advantage is practical usability and broader network support, but it depends on available liquidity, supported routes, and the assumptions of the Celer State Guardian Network.

Hyperbridge represents a more security-oriented approach. It focuses on cross-chain communication using cryptographic verification mechanisms, including consensus and state proofs. This can reduce trust assumptions and support heterogeneous blockchain communication, but it is mainly focused on protocol-level interoperability rather than simplifying the end-user workflow.

A workflow-oriented example was **TurtleV2**, developed by Velocity Labs. It aimed to simplify token movement in the Polkadot ecosystem by combining existing mechanisms into a unified user-facing process. It supported Snowbridge-based transfers between Ethereum and Polkadot and later added XCM transfers between Polkadot parachains. TurtleV2 was later discontinued and is no longer operating.

4.2.5 Summary of limitations

The reviewed solutions show that interoperability is already addressed from several perspectives. Some protocols improve transfer speed inside selected ecosystems while trust-minimized bridges improve security between heterogeneous blockchains. So the limitation is not the lack of interoperability tools, but their limited combination of transfer scenarios and use cases.

During the development of this thesis, the selected scenarios were not available as one continuous workflow covering Layer 2 to Layer 2, Layer 2 to Layer 1, Layer 1 to Polkadot, and combined Layer 2 to Polkadot transfers. The gap addressed in this thesis is application-level coordination of selected tools for continuous Ethereum and Polkadot transfers and providing guidance for the user.

4.3 Solution overview (high level)

The proposed solution is a web-based application layer for simplifying asset transfers across multiple blockchain networks. The main purpose of the solution is not to replace existing interoperability protocols, but to improve how selected transfer possibilities are presented and coordinated for the user. Protocols that exist now, already provide mechanisms for transferring assets between networks, but they are often exposed as separate tools with different interfaces, supported routes, and execution steps. This creates additional complexity, especially when the transfer cannot be completed through one direct action.

The proposed application addresses this problem by creating one guided transfer workflow. Instead of manually selecting a bridge, checking supported networks, and deciding the correct order of steps, the user describes the intended transfer through a simple set of inputs. These inputs include the source network, destina-

tion network, asset, amount, and recipient address. All information besides the recipient address are provided by the proposed application. Based on this information, the application prepares a high-level transfer plan and guides the user through the required process.

4.3.1 Main idea of the solution

The main idea is to separate the user's intention from technical parameters. From the user's point of view, the task is to simply move an asset from one network to another. However, the technical process behind this action may require one or more interoperability mechanisms. Some transfers can be executed directly, while others require an intermediate network and more than one transaction.

The application therefore works as a coordination layer. Instead of creating new mechanisms and protocols, it uses existing interoperability mechanisms and combines them into one user-facing workflow. The value of the proposed solution is in planning, simplifying, and presenting the transfer process in a way that is easier for the user to follow, while combining the advantages of both tools in use.

This approach seems most appropriate because the main difficulty is not the lack of interoperability tools. It is quite the opposite. since there are many tools focusing on different types of the transfers its harder for user to find routes which are technically possible but can be harder to execute because of the knowledge necessary to understand the right path, sequence of steps and which assets can be send to exactly which destination and also which tools supports such actions. The proposed solution reduces this complexity by moving part of the decision-making into the application layer.

4.3.2 Design logic

The design was based on the idea that creating a new interoperability protocol would require solving complex problems, including message verification, liquidity management, security of smart contracts, protocol operation and maintenance in long-term. Based on all these reasons and also the fact that many similar solutions already exist which were developed by bigger teams and fundings, the proposed work focuses on combining already existing solutions and not replacing them.

Before the implementation, available interoperability approaches were reviewed. The approach selected had to support two main types of transfers. The first type is transfers between Ethereum-based networks. The second type is transfer between different ecosystem, where the source and destination networks do not share the same architecture. Another thing to consider was that both transfers had to be working both ways. If not, the proposed solution would be incomplete and would allow transfers to be executed both ways.

4.3.3 Supported transfer scenarios

The proposed solution supports three different groups of transfer scenarios.

- The first group contains transfers between EVM-based networks. These transfers represent homogeneous interoperability because the networks share similar user interaction models, wallet behavior, and address formats.
- The second group contains transfers between Ethereum and the Polkadot ecosystem. These transfers represent heterogeneous interoperability because the two ecosystems use different architectures and different technical models.
- The third group contains combined transfers. These transfers are needed when the source network and destination network are not directly connected

through one simple step. For example, an asset may need to move from a Layer 2 network to an intermediate Layer 1 network before it can continue to another ecosystem. In such a case, the application prepares a multi-step workflow and executes the steps in the correct order.

These transfer scenarios were selected because they use the full scale of the available combinations of transfers which two selected tools can offer. In this way the proposed application has a diverse selection of different types of transfers from different sources to wider selection of destinations.

4.3.4 Workflow and architecture overview

The user selects the source network, destination network, asset, amount, and recipient address. Based on this input, the application checks whether the selected transfer can be processed and creates a transfer plan. This plan shows whether the transfer can be executed as a single step or whether it requires multiple steps. Before the execution, the transfer plan is presented to the user for quick overview. During execution, the application monitors the progress, and continues to the next step only after the previous one is completed. The user still signs all transactions through their own wallet, so the application does not take custody of funds.

4.3.5 Limitations of the proposed solution

The main limitation of this proposed system is the fact that it relays on the capabilities of the used mechanisms. If a selected asset or route is not supported, the application cannot show such combination. In this way this limitation is detected but can't be solved since it cannot create a new missing route which is not already made by one of the other tool.

The other limitation is execution time of multi-step transfers. Since the transac-

tions are dependent on the tools used the transaction times are as fast as the tools them self.

The solution is also affected by the differences between supported ecosystems, such as different wallet types and address formats. The application can make these differences easier to manage but it cannot fully remove them.

4.4 Risk assessment

Students should recognize risks attached to how they implement their solutions and offer mitigation strategies.

4.5 Experimental reproducibility and integration

The solution engineering phase should be conducted with reproducibility and systems integration in mind and students should describe how they tackled this aspect in detail. Unless special conditions apply, code, models and data used for solution engineering should be freely available. Machine learning models should be presented to allow further use without retraining. • Where possible, use docker-ized solutions ensuring product or service can be used even if the underlying technology changes.

4.6 Sustainability and environmental impact

Students should describe how would they implement measures ensuring sustainability of their product or service during lifecycle

4.7 Employability

Students should be able to reason why knowledge and skills that they have learned during bachelor work will help them to secure better position on job market.

4.8 Teamwork, diversity and inclusion

When domain experts from multiple fields of study are involved in a project, student should implement processes and describe techniques for sharing tasks and (understanding) knowledge between different parties. Problems and mitigating strategies should be described to ensure that the project is finished within the given time schedule. This part should also involve consideration on diversity and inclusion matters.

Chapter 5

Conclusion

Students should summarize key aspects of their work, considering engineering and societal impact.

Chapter 6

References

Up to 70 references.

Chapter 7

List of figures

Appendix A

First Appendix

Appendix B

Contents of Included CD-ROM

CD-ROM included to the thesis contains following files:

- `/file1` — First file
- `/file2` — Second file